

The Elements of Prompting — Field Guide

Requirement quality · elements as requirement types · menus by artifact · the five exercises, worked · the evidence compendium.

Why elements instead of “good prompts”: output quality swings up to 76 accuracy points from formatting alone (Sclar et al., ICLR 2024). You don’t reword your way to a reliable prompt — you find which element failed and fix that one. Learn the elements and you can diagnose any bad output, and build templates of your own instead of collecting other people’s.

THE DIAGNOSIS GRID

The output is...	The element that failed
the wrong altitude, tone, or posture	Role
an answer to a different (or vaguer) question	Task
generically right, specifically wrong for us	Context
right content, unusable shape or length	Format
not matching the standard in your head	Examples
confidently invented	The Out (missing)
sprawling past what you asked	The Stop (missing)

PART 1 — REQUIREMENT QUALITY

Write requirements, not wishes

You already write work specifications: requirements, standard operating procedures, acceptance criteria, test protocols. A prompt is the same craft at small scale — it specifies the content, behavior, and quality of a reply or artifact. Two ideas carry the whole course: requirement type (WHAT must be specified for this artifact — parts 2, 3, and 6) and requirement quality (how well each one is written — this page). The quality bar below applies the writing principles summarized by NASA and INCOSE; it improves clarity and checkability without guaranteeing model compliance.

Quality	For a prompt, it means	The failure it prevents
Necessary	Every line earns its place; a simple task needs only a few requirements.	Rule piles that bury the ones that matter.
Clear	One reading; named columns, defined terms, no "etc."	The model resolving ambiguity its own way, silently.
Complete enough	The consequential facts and rules are stated — not every fact.	Gaps filled with the most plausible guess.
Consistent	No requirement contradicts another; trade-offs made explicit.	The model reconciling you with yourself, its way.
Feasible	Possible with the data and tools actually supplied.	Confident output resting on data that cannot support it.
Singular	One requirement per line — separable, checkable, revisable.	Half-followed compound instructions nobody can audit.
Verifiable	An observable check exists: a number to reconcile, a behavior to exercise, a list to cover.	"Looks done" standing in for done.

Three boundary rules. A role title is a starting point — observable behavior makes it useful. Context also supplies background facts: turn the consequential ones into explicit instructions when their use matters. And prompt instructions request compliance — actual permissions and configured controls determine which actions a tool can take. **Verification** asks whether the output meets the written criteria; **validation** asks whether it serves the actual need — check both, every time.

PART 2 — GENERATIVE ELEMENTS: THE SEVEN REQUIREMENT TYPES OF A REPLY

A prompt that writes: five elements, two safety valves

A generative prompt commissions one piece of text you will read and act on. Everything you need to control fits in five elements — plus two one-sentence safety valves that shut off the two signature failures.

1 · ROLE — who is answering

Definition. The identity, expertise posture, and — critically — the behavioral commitments you assign before the task begins.

Mechanism. An LLM completes documents by predicting what plausibly comes next. The role selects which region of everything it has read the completion is drawn from: vocabulary, caution level, what gets foregrounded. “You are a careful analyst” makes analyst-style continuations more probable — it does not add analyst knowledge.

Evidence & nuance. Personas do not improve factual accuracy (162-persona study, Zheng et al. 2024) — “you are a genius mathematician” doesn’t fix arithmetic. They do reliably shift tone, framing, and foregrounded knowledge (Salewski et al. 2023). Use Role for how it answers; never as a substitute for checking what it answers. Behaviors beat titles because behaviors are auditable in the output.

Weak: “You are a senior data scientist.” — a credential; produces nothing checkable.

Strong: “You are a careful analyst. You compute every number by running code, you state n for every proportion, you never drop data silently, and you say ‘the data can’t answer that’ when it can’t.” — four behaviors, each auditable.

Failure it prevents: the generic-assistant voice — mid-distribution, hedge-everything output.

2 · TASK — what, for whom, to what end

Definition. The single clear ask: a verb + object + audience + success criterion.

Mechanism. The most load-bearing words in the prompt — they define the target the completion optimizes toward. Vendors agree to the point of redundancy: Google calls the task “the most important component”; for Microsoft the Goal is the only required part.

Nuance. One task per prompt; if several, number them in priority order and add the Stop. A question beats an instruction because it carries its own completion test — either it got answered or it didn’t. Include the audience and the outcome: “the reader should be able to decide X.”

Weak: “Analyze the returns data.” — a topic. Topics generate exploration.

Strong: “Answer: what is the return rate by site for Q2 versus the 2.0% target, and which three reason codes drive 80% of returns?” — questions generate answers.

Failure it prevents: wandering output — ten observations and no answer.

3 · CONTEXT — what the model cannot know

Definition. Everything true in your world that isn’t in the training data: raw material, definitions, constraints, history, quirks — and the reasons behind your rules.

Mechanism. The model fills every information gap with the most statistically plausible filler. Context replaces plausible-in-general with true-for-you — this is the anti-hallucination element; most “lies” are really unsupplied context.

What goes in it. The raw material itself (never assume it knows your documents) · a glossary for terms with local meaning, used exactly · constraints and sensitivities · known quirks each with the rule to apply · and the why behind non-obvious rules — “this will be read aloud, so no ellipses” measurably outperforms “no ellipses”; models generalize from reasons.

Discipline. Delimit context from instructions (tags, fences, headers) so material to work on is never confused with orders to follow — also your first defense when the material itself contains instruction-like text.

Weak: “Use our standard definitions.” — the model’s “standard” is the internet’s average.

Strong: “Glossary — use exactly: Return rate = returned units ÷ shipped units (shipped_q2.csv). Known quirk: the export has a totals row at the bottom — exclude it and say so.”

Failure it prevents: generically-correct, specifically-wrong output — the plausible number with your company’s name on it.

4 · FORMAT — the output contract

Definition. The required shape of the response: structure, length, tone, medium, what to include and omit.

Mechanism. Models imitate structure more readily than they obey descriptions of it — so Format works best stated positively (“respond in flowing prose paragraphs” beats “don’t use markdown”) and best of all demonstrated (see Examples).

Nuance. Give a numeric length cap with a priority rule (“≤120 words — cut content before quality”); adjectives like “short” are unenforceable. On genuinely hard reasoning, heavy format constraints can tax accuracy — let it reason free-form first, then format the answer in a second step.

Weak: “Keep it short and professional.”

Strong: “Max 120 words. Structure: the ask in sentence one; two supporting facts with numbers; the deadline. Nothing else.”

Failure it prevents: the 3×-too-long draft; the buried answer; output you must re-shape by hand.

5 · EXAMPLES — show, don’t describe

Definition. 3–5 realistic demonstrations of the task done right, placed in the prompt.

Mechanism. In-context learning (Brown et al. 2020 — the discovery that founded prompt engineering): the model infers the pattern from demonstrations without retraining. The strongest steering signal that exists for format, tone, and edge-case handling.

Nuance. Diversity is the point: include the awkward case (missing field, borderline grade, angry customer) — the model handles edges exactly as your examples do. Wrap them in tags so they can’t be mistaken for live input. 2026: on thinking models, start zero-shot; add examples only when format drifts (few-shot measurably degrades some reasoning models).

Weak: three examples that are the same happy case reworded.

Strong: one typical case + one edge case + one reject/escalate case — now the model knows the boundaries, not just the center.

Failure it prevents: the model guessing your standard from the internet’s average; format drift across runs.

THE TWO SAFETY VALVES

THE OUT — permission to not know. “If the document doesn’t say, say so.” Models are trained on benchmarks that reward guessing over abstaining (OpenAI, 2025); the Out re-opens the abstain option and drastically cuts invented answers. Without it you have implicitly ordered the model to always produce something.

THE STOP — the scope boundary. “Answer these three questions, then stop — do not go exploring.” Converts open-ended capability into bounded work. In agentic prompts it grows up to become gates and autonomy rules.

PART 3 — AGENTIC ELEMENTS: REQUIREMENTS FOR DELEGATED WORK

A prompt that works: the twelve blocks of the mission brief

An agentic prompt commissions a job, not a text: the agent plans, acts through tools, checks results, and iterates — mostly while you are not watching. Every generative element still applies. The additional blocks exist for one reason: text that fails costs you a re-prompt; actions that fail change the world — files overwritten, emails sent, wrong numbers published. So the agentic elements govern conduct: where the agent may act, how it must verify itself, when it must stop and ask, and how it proves what it did. (Template A1; worked version A2.)

1 · <role> — the standing behavioral contract

Same element as generative Role, with higher stakes: in chat it shapes one answer you’re about to read; in an agent it biases hundreds of unsupervised decisions. Write it as the behaviors you’d want if you couldn’t check any single step: “never invents numbers, logs every transformation with a row count, prefers reproducible scripts over manual edits.” **Prevents:** an agent that improvises its professional standards mid-run.

2 · <mission> — the goal and the definition of done

The agentic Task: one goal, the decision it feeds, audience, deadline — plus the line chat never needs: “done looks like: ...”. An agent runs a loop; the mission is its exit condition. Vendor grounding: give the agent “a check it can run... without one, ‘looks done’ is the only signal available.” **Prevents:** the agent deciding for itself what finished means — plausible activity, indefinitely.

3 · <context> — tribal knowledge, written down

Same element as generative Context; the agentic difference is coverage pressure — an agent touches your domain at every step, so every unstated quirk gets stepped on. The glossary becomes a controlled vocabulary (it will otherwise compute three subtly different “yields”); each quirk carries its rule (“Site B records this part under a legacy code — map it to the canonical number”). **Prevents:** silent invented fixes — the most expensive agent error, because they look like diligence.

4 · <environment> — the workspace contract

New in kind. Where the agent may read, where it must write, what tools exist, what is forbidden. Load-bearing distinctions: inputs are read-only · outputs are versioned (never overwrite what a human opened) · scratch is regenerable. Name forbidden actions explicitly (no network calls, no emails, no writes outside the folder). Least privilege as prose — enforce with sandboxing too: the prompt is policy, the sandbox is physics. **Prevents:** the agent “helpfully” editing your master workbook.

5 · <inputs> — the source register

New in kind. One block per source: location, exact format, grain (“one row = one inspected unit” vs “one lot with counts” — the line that determines every denominator downstream), primary key, coverage, owner, trust level (authoritative / derived / manual entry), known issues. Cardinal rule: a fact you can’t establish is written **UNKNOWN** and raised as an open item — never guessed. **Prevents:** the agent inferring your schema — where most wrong numbers are born.

6 · <plan> — bounded method

The steps in order, plus two lists chat prompts don’t carry: methods allowed (boring, reproducible) and methods not allowed without asking (dropping data, causal claims, model fits, new dependencies). Specific enough to prevent improvised rigor, open enough to adapt to what it finds. **Prevents:** creative methodology at step 14 of 20, discovered only at review.

7 · <checks> — machine-evaluable verification

New in kind — the honesty engine of the whole brief. Verification the agent can actually run, split by consequence: HARD checks stop the run (totals reconcile to a trusted number; pass + fail = inspected; no duplicate keys); SOFT checks flag and continue (null spikes, small n, metric moved >X vs last cycle). Two disciplines: write checks as equalities and set-memberships, not aspirations (“check data quality” is not a check); and always include one reconciliation anchor — a number you already trust from outside the run. Without an anchor, every other check can pass on the wrong data. **Prevents:** plausible-but-wrong deliverables — the agent’s version of hallucination.

8 · <outputs> — the deliverable contract and audit trail

The agentic Format: exact filenames (with date suffixes), exact locations, and the three logs that make a run auditable — CHANGES.md (what changed, when, why) · FINDINGS.md (claims → evidence → confidence) · OPEN_ITEMS.md (assumptions and questions, each with owner and status). A colleague should be able to find, understand, and regenerate everything from the folder alone. **Prevents:** results you can’t locate and conclusions you can’t trace six weeks later.

9 · <process> — phases and human gates

New in kind. The job cut into phases with human checkpoints at the cheap-error points: Gate 1 confirms definitions before any data is touched · Gate 2 reviews reconciliation before analysis builds on the numbers · Gate 3 approves headlines as plain text before anything renders or ships. A definition error caught at Gate 1 costs a sentence; the same error after the deck circulated costs a retraction. **Prevents:** compounding — hours of competent work on a wrong foundation.

10 · autonomy rules — when to proceed, when to stop (inside)

The agentic Stop, matured. Between gates the agent proceeds without asking; it must stop when: a HARD check fails · an input isn’t as described · a definition is ambiguous · an action is hard to reverse (delete, send, publish, overwrite) · a previously published number would change. Both halves matter: without “proceed” you built a slow chatbot; without “stop” you built a liability. General principle: reversible → act; irreversible → ask.

11 · rules + quality_bar — the invariants, and done-as-a-checklist

<rules> are the non-negotiables kept short and absolute so they survive a long context: never invent or back-fill a number · sources are immutable · every transformation logged with a row count · assumptions stated as assumptions. Rules differ from checks: checks are evaluated at defined points; rules hold everywhere. **Prevents:** standards drift in hour three.

<quality_bar> is the definition of done made auditable: every placeholder resolved or logged · all HARD checks pass, every SOFT flag noted · every deliverable re-opened and verified · logs written · every presented number traceable to a source cell. Checks verify the work mid-run; the quality bar verifies completeness at the end. **Prevents:** “done” meaning “I stopped.”

12 · <reporting> — the interface back to you

The agentic Format for the message, fixed in shape: status (done / stopped at gate n / blocked) · headline answer with the number · deliverable paths · checks summary · open items needing a human · risks. Under 25 lines; evidence, not assertions — the test it ran and what it returned, not “everything went well.” Details live in the files. **Prevents:** a narrative essay where a status should be.

PART 4 — THE MAPPING

The agentic brief is the generative anatomy, grown up to survive autonomy

Generative element	Agentic descendant(s)	What was added, and why
Role	<role>	persists across unsupervised decisions → behaviors only
Task	<mission>	+ definition of done — the loop needs an exit condition
Context	<context> + <inputs>	+ per-source register with grain & trust — agents touch every source repeatedly
Format	<outputs> + <reporting>	+ audit logs and a fixed status shape — actions must be traceable
Examples	the filled brief (A2)	a worked run is the agentic few-shot
The Out	UNKNOWN → open items	not-knowing becomes a logged, owned item
The Stop	gates + autonomy rules	scope control becomes checkpointed process
(none)	<environment> <plan> <checks> <rules> <quality_bar>	new in kind — conduct, method, verification, invariants, completeness: things text-only prompts never needed

PART 5 — BUILDING YOUR OWN TEMPLATES

A template is elements + reasons, assembled against failure modes

1. Name the task’s three most expensive failure modes (wrong denominator? invented facts? unusable format? irreversible action?).
2. Pick the element that owns each failure — the diagnosis grid on page 1; reversibility → autonomy rules.
3. Write those elements strong; keep the rest light. A template where every element is maximal is a template nobody fills in.
4. Add one filled example — the gold standard teaches faster than the instructions do.
5. Test on 3–5 real cases (one edge case), version it, name an owner, add it to the library.

PART 6 — REQUIREMENTS BY ARTIFACT

The menu, per artifact: pick what resolves ambiguity

The artifact decides which requirement types matter: calculation rules for a spreadsheet, behavior and state for a dashboard, narrative and editability for a presentation, chronology and evidence for an email brief, reading level and fidelity for a document. For each selected type write: requirement → example → acceptance evidence. These menus are maintained in Requirements_by_Artifact.md (the shared catalog behind the Template Creator and the Configurator); this is a teaching taxonomy, not a formal standard.

SPREADSHEET ANALYSIS

Requirement type	Example instruction	How to check it
Row grain	State that each detail row is a month–site–supplier observation.	Confirm grouping keys before aggregation.
Reporting period	Use an explicit inclusive month range from Month.	Check the first and last included months.
Field types and units	Treat Month as a month; identify counts, percentages, hours, and currency.	Check parsing and unit labels.
Duplicates and summary rows	Exclude the existing TOTAL row and investigate duplicate keys.	Compare row counts and a duplicate-key report.
Metric definition	Define return rate as sum of Units_Returned divided by sum of Units_Shipped.	Recompute the numerator, denominator, and displayed rate.
Weighting and denominators	Do not infer overall on-time delivery from monthly percentages without delivery counts.	Flag the missing denominator instead of claiming an overall rate.
Missing and invalid values	Retain the missing Inspection_Hours value; disclose any exclusion.	Inspect missing-value counts and affected calculations.
Segmentation and comparison	Compare suppliers on the same selected period and sites.	Confirm equal filter scope for each comparison.
Rounding and number formats	Display return rates to three decimal places while retaining calculation precision.	Check displayed 0.348% against the underlying ratio.
Formula traceability	Use inspectable formulas or an included calculation log.	Trace one output to its input rows.
Structure and usability	Provide a scorecard tab and a definitions tab with meaningful headers.	Open both tabs and identify the measure without outside explanation.
Verification and limitations	Reconcile counts and sums; distinguish association from a causal explanation.	Compare totals and challenge unsupported explanations.
Delivery and portability	Deliver an editable Excel workbook (.xlsx) with no broken external references.	Open the file in the intended spreadsheet application.

INTERACTIVE DASHBOARDS

Requirement type	Example instruction	How to check it
Input contract	Embed the checked supplier detail data and identify the snapshot date.	Compare record count and source identifiers.
Date range	Use month selectors matching the source granularity; include both endpoints.	Test a single month and the full range.
Category controls	Offer Supplier and Site multi-select controls; Type or Class requires a real source field.	Compare each option with actual distinct values.
Grouping and granularity	Allow grouping by supplier, site, or month when relevant to the decision.	Check counts and totals after changing the grouping.

Requirement type	Example instruction	How to check it
Metric selection	Offer named measures with definitions and units.	Verify that labels, scales, and calculations all change together.
Filter combination	Apply selected filters consistently, with an explicit AND/OR rule.	Use a known intersection and compare included records.
Coordinated views	Charts, totals, and tables must represent the same selected records.	Reconcile a chart total with its underlying table.
Sorting and drill-down	Sort numerically and allow inspection of supporting rows.	Compare sorted order and inspect a selected group.
Empty, missing, and zero states	Distinguish no matching records from a valid zero; avoid division by zero.	Exercise all three cases.
Reset and visible state	Show active selections and restore the default state with Reset.	Change multiple controls, reset, and compare the initial view.
Accessibility and layout	Use labeled controls, visible focus, readable contrast, and a usable narrow layout.	Navigate by keyboard and test a smaller window.
Portability and dependencies	Deliver one self-contained HyperText Markup Language (.html) file with embedded data and no required network calls.	Open offline and inspect for missing assets.
Performance	Choose a measurable response target on the training laptop for this dataset.	Time a representative filter change; do not invent a universal target.
Export and reconciliation	If export is needed, export the filtered data and retain the chosen scope.	Compare exported rows with the displayed selection.

PRESENTATIONS

Requirement type	Example instruction	How to check it
Purpose and audience	Name the decision, audience knowledge, and intended action.	A reviewer can identify the decision and next step.
Scope and slide count	For the mock presentation, request five slides with a defined coverage list.	Count slides and map each required topic.
Story structure	Move from business question to evidence, implications, and next action.	Read only the titles and check the narrative.
Message hierarchy	Use one main message per slide; make evidence support that message.	Ask a reader what to remember first.
Data fidelity	Use only checked findings and preserve period, units, and limitations.	Reconcile displayed values with the workbook.
Chart semantics	Choose an appropriate chart, truthful axes, and clearly labeled categories.	Check that the encoding supports the claimed comparison.
Typography and readability	Set readable projected text and avoid dense paragraphs on teaching slides.	Inspect at presentation size and the expected viewing distance.
Visual consistency	Use a restrained palette, aligned margins, and consistent emphasis.	Compare consecutive slides as a set.
Speaker notes	Provide purpose, talking points, transitions, and exercise instructions.	Inspect notes on every relevant slide.
Timing and mode	Classify TEACH, DO, REFERENCE, and DIVIDER; protect 30 practice minutes.	Sum notes and rehearse the live path.
Reference completeness	Make self-study slides understandable without the facilitator.	Have a reader explain the example from the slide and notes.

Requirement type	Example instruction	How to check it
Accessibility	Use readable contrast and descriptive alternatives for meaningful visuals.	Review with the target presentation application's accessibility tools.
Delivery and editability	Deliver an editable PowerPoint (.pptx), retaining notes and usable charts.	Open, edit, and present the exported file.

EMAIL SUMMARIES

Requirement type	Example instruction	How to check it
Source scope	Name the selected thread, relevant date range, and available attachments.	List the messages and attachments actually used.
Current state	Separate the latest position from earlier proposals.	Trace each changed date to the message that superseded it.
Decision extraction	Distinguish approved, proposed, rejected, and unresolved items.	Compare decision status with the source wording.
Conditional approval	Keep the cost cap and quality sign-off attached to the approval.	Reject a summary that says simply approved.
Actions and ownership	Extract action, explicit owner, due date, and status.	Check each field; use Not stated for missing details.
Date meaning	Distinguish a trial date, change date, and action due date.	Check that September 25 and October 2 are not conflated.
Open questions and conflicts	List unresolved freight responsibility and contradictory statements.	Locate the source question and any subsequent resolution.
Evidence traceability	Use message dates, senders, and supporting excerpts or available links.	Open the referenced message and verify the claim.
Attachments and missing input	Flag a mentioned drawing that is not available to the assistant.	Do not summarize content of an unavailable attachment.
Format and tone	Provide a neutral current-state brief and a separate action table.	Check that opinions are labeled and actions are easy to scan.
Access and tool availability	Use only mail and attachments available in the actual Copilot and Outlook setup.	Confirm retrieval scope; use the supplied-text exercise when needed.
Authority and delivery	Prepare any reply as a draft for human review; do not send automatically.	Inspect the draft and action record.

PLAIN-LANGUAGE DOCUMENTS

Requirement type	Example instruction	How to check it
Reader and purpose	Write for a reader at a fifth-grade reading level who needs to understand or do a named task.	A representative reader can state the purpose.
Source and factual scope	Use the supplied source and identify unsupported or missing information.	Trace factual claims to the source.
Concept coverage	List the ideas the reader must understand before drafting.	Compare the document with the coverage list.
Vocabulary	Prefer familiar words and define necessary technical terms at first use.	Review unfamiliar terms and their definitions.
Sentence and paragraph structure	Use short direct sentences and one main idea per paragraph.	Read aloud and inspect complex sentences.
Organization	Use meaningful headings and ordered steps when order matters.	Ask a reader to locate the first action and a key explanation.

Requirement type	Example instruction	How to check it
Examples and analogies	Include a concrete everyday example and state the limit of any analogy.	Check that the analogy does not change the underlying idea.
Tone and respect	Use a clear adult tone without condescension or childish phrasing.	Review wording with the intended audience.
Accuracy and caveats	Preserve important conditions, uncertainty, exceptions, and warnings from the source.	Compare the simplified version with the original meaning.
Readability evidence	Use reading-level estimates as one signal, plus human comprehension checks.	Do not claim exact grade-level comprehension from a score alone.
Visual support and accessibility	Use a labeled diagram or example only when it improves understanding.	Ask what the visual helps the reader explain.
Comprehension check	Provide three questions and a short answer key grounded in the document.	Confirm the answers can be found or inferred from the text.
Delivery and reuse	Specify an editable document, useful headings, and reusable topic placeholders.	Open the output and adapt one topic without losing the structure.

PART 7 — THE FIVE EXERCISES, WORKED

The course prompts — copy, adapt, keep the checks

One fictional supplier case runs through the first three: analyze → dashboard → presentation, on the same cleaned records and definitions. Full step-by-step versions with iteration commentary: Course_Workbook tabs 1–5. Every prompt below is designed to be copied straight into an assistant.

1 • Analyze spreadsheet data

Use Course_Workbook.xlsx, Data tab, detail rows only – exclude the TOTAL row (144 rows; one Inspection_Hours cell is "n/a": treat it as missing, not zero, and disclose it). One row = one month × site × supplier; period 2025-09 to 2026-08 inclusive. Question: which supplier has the highest RETURN RATE, defined as total Units_Returned ÷ total Units_Shipped per supplier over the full period – computed from sums, never by averaging monthly percentages. Keep Defects_Found separate from returns; do not add them together. Rank the three suppliers showing numerator and denominator, reconcile total shipped and returned against the TOTAL row, and state one limitation of the comparison. Then stop.

Acceptance evidence. Check: 144 detail rows · 224,902 shipped · 432 returns · highest return rate Bravo Plastics ≈ 0.348% (262 ÷ 75,184) · reconciles to the TOTAL row · defects (2,207) reported separately.

2 • Build an interactive dashboard

From the attached data file, build ONE self-contained HTML file that works offline from disk. Views: summary tiles, one chart, and a sortable table of the records – data embedded, every number computed from the embedded records, metric definitions and the snapshot date in the footer. Controls, wired so one selection re-scopes every view: a month-range selector with inclusive endpoints · category multi-select filters · a comparison switch · a monthly/quarterly granularity switch · a metric selector · drill-down from a chart segment to its underlying records · ranking on the selected measure · Reset. Show active selections; an empty selection says "no matching records"; missing values stay blank; a zero denominator shows "-". Label the file a static data snapshot with its date.

Acceptance evidence. Check: Berlin × Bravo Plastics, 2026-03..2026-08 → 6 records, 8,575 units, 21 returns, 0.245% — identical in tiles, chart, and table. Prepared output: Supplier_Quality_Dashboard.html (footer self-check).

3 · Prepare a presentation

Create an editable five-slide management presentation using only the verified analysis. The audience must decide what follow-up is warranted. Cover: the business question and scope · the comparison · one useful time or segment view · a proposed follow-up · limitations with next steps. Give each slide one main message, readable evidence, accurate units and reporting period, and brief speaker notes. Preserve uncertainty and missing-data limits. Do not invent causes or unsupported benefits. Check every number against the analysis and inspect the exported slides for clipping and readability.

Acceptance evidence. Check: five slides mapped to the coverage list · every number reconciled · findings separated from recommendations · exported slides inspected. Prepared output: Supplier_Quality_Mock_Presentation.pptx.

4 · Summarize an email conversation

Review the selected conversation about [topic] using only the messages and attachments you can access. Create a brief of the current situation, followed by tables of decisions and actions. For each decision: status, conditions, supporting message. For each action: task, explicitly stated owner, due date, status, supporting message. Distinguish current dates from superseded dates. List unanswered questions, conflicts, missing information, and referenced attachments you cannot access. Preserve approval conditions. Write "Not stated" for missing details. Separate source facts from your interpretation. End with the next clarification needed. If requested, draft a reply for review; do not send it.

Acceptance evidence. Check (the four traps): change date Sep 25 → Oct 2 conditional on sign-off · trial date Sep 25 supersedes Sep 18 · 18,000 cap holds, freight unresolved · drawing referenced but not accessible. Source: Packaging_Change_Thread.txt.

5 · Explain a topic clearly

Using [source], explain [topic] so a reader at a fifth-grade reading level can understand it. State the main idea first. Use familiar words, short sentences, meaningful headings, and one concrete example. Define necessary technical terms when first used. Use an analogy only if it is accurate, and explain where it stops being useful. Preserve important conditions and uncertainty. Do not invent facts or use a childish tone. End with three comprehension questions and a short answer key. Check the explanation against the source and identify any simplification that changes the meaning.

Acceptance evidence. Check: main idea first · terms defined at first use · analogy limit stated · conditions preserved against the source · three answerable questions · adult tone. Samples: exercise-data/plain-language/.

PART 8 — THE EVIDENCE COMPENDIUM

What works, what's myth, what expired — and why

Half the prompting advice in circulation is folklore from 2023. This chapter is the course's evidence layer in one place — every technique given a verdict, with the why behind it. Three verdicts: **WORKS** (replicated — use it) · **MYTH** (never survived replication, including some standard 2023 advice) · **EXPIRED** (was genuinely right, then the product absorbed it). Researched September 2026 — this field moves; the study behind every line is in the PLAYBOOK tab of the Course Workbook and [notes/research/](#).

WORKS — replicated, reliably helps today	
Be specific: task, constraints, success criteria. The model optimizes toward the target you state — every unstated need is filled with the most plausible average guess, which matches your intent only ~41% of the time, and vague prompts double the risk of a wrong turn.	Yang 2026 (41.1%)
One tested, versioned template — delimiters separating instructions from material. Models are exquisitely sensitive to wrapper and layout: formatting alone swings accuracy by up to 76 points, the wrapper alone by up to 40%. A tested template removes that lottery and makes runs comparable.	Sclar, ICLR 2024 · He 2024
Long inputs: documents at the top, question at the END — bookend both ends when very long. Attention concentrates at the edges of the context and thins in the middle (“lost in the middle”); the last thing read is the freshest instruction. Placement alone is worth up to ~30% on long documents. (The 2023 recipe — “instructions, ###, then the text” — was written for 4-8K-token windows; as contexts grew to a million tokens the guidance inverted.)	Liu 2023 · Anthropic docs · GPT-4.1 guide 2025
Give it an out — in checkable shape — and require citations. Models are trained on benchmarks that reward a confident guess over “I don’t know,” so inventing is the default way to obey you. The out re-opens honest abstention; the checkable shape (“anything not stated: write UNKNOWN — never estimate; list the gaps”) makes compliance visible in the output.	Omar 2025 (66%→44%) · OpenAI 2025
Zero-shot first; add 3-5 targeted, diverse examples only when format or tone matters. Examples are the strongest format signal there is (in-context learning) — but modern reasoning models degrade under example piles. Demonstrate the standard, edge case included; don’t crowd the reasoning.	OpenAI reasoning practices · Anthropic 2026
Concrete numeric budgets for anything measurable — words, bullets, steps, tool calls. A number is enforceable and self-checkable; an adjective (“short,” “concise”) is a mood the model resolves to the internet’s average. Current models adhere well to concrete length guidance.	GPT-5.1 guide · GPT-5 guide
Persona and audience for VOICE and level — never for accuracy. A role selects which region of everything the model has read the answer draws from — vocabulary, caution, framing. It adds no knowledge: assigned traits are detectable in output up to 80% of the time while accuracy doesn’t move.	PersonaLLM 2024 · Google PTCF
Self-check against NAMED criteria, with evaluation blinded from generation. “Check your work” produces a generic pass. Named criteria give the check something falsifiable to fail, and blinding the checker from the draft’s authorship stops the mirror from grading its own reflection. Fact scores jump 56→71 with planned verification.	CoVe, ACL 2024 · Cheng, Science 2026

Ask the AI to improve your prompt (metaprompting). Prompt wording is a search problem, and models search phrasing-space better than people do — optimizer-written prompts beat human ones, and every major vendor now ships an official prompt improver. Describing what you want and letting the AI draft the prompt is a proven move, not cheating.	OPRO, ICLR 2024 · GEPA 2026 · official vendor tools
MYTH & MISTAKE — never worked, or hurts today	
“Tell it it’s a genius and it gets smarter.” Personas move style, not competence — knowledge is not gated behind flattery. 162 expert personas tested: zero accuracy gain; dumbed-down personas actively hurt. Standard 2023 advice that never survived replication.	Zheng, EMNLP 2024 · Wharton R4, Dec 2025
Tips, threats, “this is important to my career.” No stable mechanism: effects wash out on average while adding $\pm 35\%$ per-question chaos. The famous “+115%” was the cherry-picked best case — the honest average is $\sim 2.6\%$, null on modern models. Real stakes stated as factual context (“this goes to a regulator”) are fine: they change the task, not the mood.	Wharton R3 · Salinas 2024 · EmotionPrompt recalc
Magic phrases (“take a deep breath”). Each was one optimizer’s local optimum — for one model, on one benchmark; none replicate across models (“the only trend may be no trend”). The method that found the phrase — automated optimization — is what actually works.	OPRO, ICLR 2024 · IEEE Spectrum, Mar 2024
“Longer prompts are better.” The useful variable is information density, not length. Length without information dilutes attention: reasoning accuracy falls $0.92 \rightarrow 0.68$ with $\sim 3,000$ tokens of padding. Detail helps; filler hurts.	Levy et al., ACL 2024
“Are you sure?” as verification. Pushback is a social signal, not a re-derivation — the model re-reads you, not its work, and folds: answers flip in 58% of cases, 14.7% of them right$\rightarrow$wrong. Verification is a named-criteria check or an outside source, never social pressure.	SycEval, AIES 2025 · Huang, ICLR 2024
“The AI knows what I mean.” It fills every unstated requirement with the most statistically plausible filler — right $\sim 41\%$ of the time. Deliberately the same number as WORKS №1: “say what you mean” is the cure for this myth.	Yang 2026
Bare “do not hallucinate” commands. A ban with no honest alternative triggers over-refusal — the model starts refusing facts that ARE in the context (researchers call it the “Safety Tax”). Say what TO do instead: the out, plus required citations.	arXiv:2601.02023, Jan 2026
Contradictory instructions. Reasoning models burn their thinking budget reconciling you with yourself — and settle the conflict silently, their way.	GPT-5 guide 2025
Piling micro-rules. Instruction-following degrades past ~ 150 simultaneous rules, biased toward the earliest — later rules quietly fall off. Ship the smallest prompt that preserves the contract. (Task detail is information density and helps; rule count is what hurts.)	IFScale · GPT-5.5 guide
ALL-CAPS / ALWAYS / NEVER as an emphasis crutch. Absolutes lock judgment in cases you didn’t foresee — numeric caps constrain quantity, which is different and fine. Reserve absolutes for true invariants.	GPT-5.x guides · Gemini 3 guide

Demanding JSON or schemas a human will just read. Structure serves whoever consumes the output; when that's a person, the schema buys nothing. When a system consumes it, use structured outputs — and let the model reason first, format second.	OpenAI 2024
Concluding “the model can’t do X” from one phrasing. Formatting alone swings results up to 76 points — one failed phrasing is a data point about the prompt, not the capability ceiling. Rephrase before you conclude.	Sclar, ICLR 2024 · Wharton R1

EXPIRED — was right in 2022-23; do the replacement instead

“Let’s think step by step” → pick a reasoning model, set the effort dial. It was real — it took one math benchmark from 10% to 41% in 2022. Then vendors built the stepping in, and manual chain-of-thought now adds latency and variability on frontier models (it remains legitimate on small local ones). Advice expires when the product absorbs it.	Kojima 2022 · OpenAI guidance · Wharton R2
Piles of few-shot examples (10+) → zero-shot first, then 3-5 format-definers. In-context learning founded prompt engineering in 2020; instruction tuning then absorbed the bulk of it, and example piles now actively degrade reasoning models.	Brown 2020 · DeepSeek-R1 paper 2025
Hand-run voting and micro-decomposed steps → effort settings; chain only for auditable intermediates. Sampling-and-voting and step-scripting were absorbed into test-time compute, and over-prescribing steps now underperforms outcome-first prompts — give a clear destination, let it choose the path. Chaining stays a feature where regulated work needs auditable intermediates.	Wang, ICLR 2023 · Anthropic 2026 · GPT-5.5 guide
Carrying your 2023-24 prompt stack onto each new model → re-baseline on every upgrade. Legacy scaffolds tuned to an old model’s quirks actively degrade newer ones. Strip to the anatomy, re-test your gold cases, re-add only what earns its place — the improvement loop (PDCA), applied to model upgrades.	GPT-5.5 guide, Apr 2026

THE BIAS UNDER THE MYTHS — THE MIRROR. AI agrees with you about half again as often as a person would (+49%; 11 leading models — Stanford/CMU, Science 2026). Root cause: trained on human preference data, and we prefer agreement. This is why “are you sure?” fails and why revealing your preferred answer poisons a review. **Countermeasures, usable today:** never reveal your preferred answer when asking for judgment · ask for the case AGAINST (“three weakest points”) · paste your own draft as “a colleague’s” · treat agreement under pushback as noise, not confirmation. **See it yourself:** rate the same paragraph in two chats — “a colleague wrote this” vs. “I wrote this and I’m proud of it” — and watch the score move.

WHY THE RULES DON'T CONTRADICT — five seams, resolved:

Caps vs. absolutes: numeric caps constrain QUANTITY; ALWAYS/NEVER locks JUDGMENT — reserve absolutes for true invariants.

Bookending vs. no-repetition: repeat verbatim at both ends of LONG context, or not at all — paraphrased blanket repetition is the anti-pattern.

Task detail vs. rule-piling: detail is information density (helps); piling is rule COUNT (past ~150, quality sags and early rules win).

Chain-of-thought's dual status: expired on frontier reasoning models only — still legitimate on small, local, non-thinking models.

The out vs. “don’t hallucinate”: the out IS “don’t hallucinate” said positively — itself a live demo of “say what TO do.”

Contested — honest edges of the evidence: politeness engineering is inconsistent in both directions (shifts up to ~11% on quality rubrics, direction depends on model and language — be normally civil, don’t engineer tone). “Prompt engineering is dead” is a half-myth: the job title faded, but phrasing still swings results by double digits on frontier models and all three major vendors shipped new prompting guides in 2025–26 — the skill moved into everyone’s job description.

PART 9 — TROUBLESHOOTING: THE FAILURE MODES

Hallucination’s four characters — and the run-time grid

The mechanism: models are optimized for the most plausible next token, not the most true one — trouble concentrates where training data is thin (rare facts, citations, numbers, names). Benchmarks reward confident guessing over “I don’t know,” so models learn to be good test-takers; the better word is confabulation — gaps filled with plausible material, in-format, so fake citations LOOK like citations. Models trained to say “I don’t know” produce roughly 3× fewer false claims. **Fluency is not evidence.** And none of the defenses below eliminates hallucination — they convert unverifiable claims into verifiable ones.

The character	What it looks like	The defense
THE CONFIDENT GUESS invented facts	Free recall where data was thin — a wrong fact, delivered with total certainty.	The out, in checkable shape, plus required citations; numbers from code, current facts from search — never free recall.
THE FAKE RECEIPT invented sources	Citations, cases, book titles that look perfectly real — and don’t exist.	Click through before quoting anywhere formal — a citation is a claim, not proof.
THE JOKE TAKEN SERIOUSLY wrong source, confident answer	Fluent, cited — and built on satire, a joke, or the wrong revision.	Ask where a claim comes from; ground the answer in documents you supply.
GARBAGE IN, GOSPEL OUT runs with your false premise	It assumes your prompt is true: feed it a wrong “fact” and it politely builds on it.	State uncertain premises as questions; ask it to check the premise first — premise-checking measurably cuts agreement bias.

THE RUN-TIME DIAGNOSIS GRID — WHEN THE RUN GOES WRONG

The diagnosis grid on page 1 names the failed element when the answer is wrong. This is its agentic twin: when a long or tool-using run goes wrong, name the context failure — then apply its cure.

Failure	The smell	The cure
POISONING	One bad “fact” entered the log early — every later step politely builds on it.	Fresh session; re-load only what you trust — the standing brief plus verified files.
DRIFT	A long run wanders off the goal; the middle of a huge context gets skimmed.	Re-inject the mission; compact (the handoff move); stop conditions in the brief.
CONFUSION	Too many tools on the desk — it picks the wrong one, or dithers between them.	Fewer tools per phase; the <plan> names the methods allowed.
CLASH	Two sources disagree and the run silently picks one of them.	One source of truth per fact, named in <inputs>; an explicit override order.

“Why did it ignore me?” Instructions stack in layers — system prompt · project file (CLAUDE.md / AGENTS.md) · skill · your message — and conflicts resolve by precedence; the layer you typed is the easiest to lose. Find which layer said what before you blame the model, and put one-job rules in a skill, not another standing paragraph. Tool descriptions are prompts too — the agent reads them like a junior reads an API doc.

PART 10 — CHOOSING THE ENGINE AND THE VEHICLE

Fast vs. thinking, the model spectrum, and chat vs. workflow vs. agent

Every vendor ships a fast tier — one pass, instant, cheap — and a thinking tier that drafts, checks and revises internally before answering (it isn’t literally “thinking”: it generates intermediate tokens that improve the final answer). The bill: thinking is charged as output tokens, the expensive kind — a hard question can quietly cost 5–20× a simple one, and deep-research modes run order-of-magnitude 100×+. The discipline: pick the smallest mode that can succeed; escalate on failure, not by default.

Think ON: multi-step analysis · math · root-cause work · code · tradeoffs across a long document — anywhere being wrong is expensive.

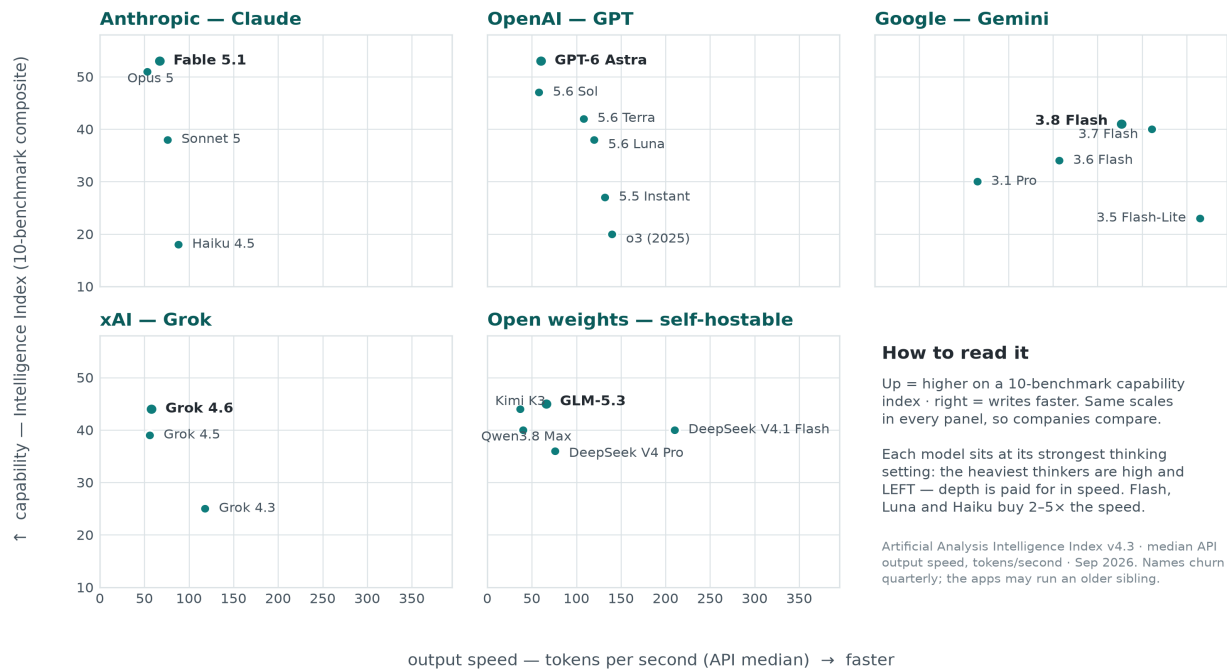
Think OFF: lookups · reformatting · summaries · routine drafting — deliberation you don’t need, at 5–20× the price.

PICK THE VEHICLE — CHAT, WORKFLOW, OR AGENT

CHAT you are the loop	Exploring, drafting, judging as you go. You read every answer and steer every turn — the generative craft, live.
WORKFLOW the loop is frozen	Known path, few tools, same steps every time — script it (n8n, Zapier, Power Automate). Cheaper, predictable, auditable; no judgment needed mid-run. A pipeline can also enforce output schemas at the system level.
AGENT the loop runs inside	Unknown path — the next step depends on what it finds. Brief it (Part 3 of this guide), with gates where the judgment stays yours. Your control lives in the brief, not the conversation.

The decision rule: known path + few tools → workflow. Unknown path, real judgment → agent, with gates. Still thinking it through → chat. **The wrong vehicle costs:** an agent on a known path pays judgment prices for clerk work; a workflow on an unknown path is a script that breaks on the first surprise. And don’t chat with an agent — hand it a work order.

The two speeds, measured — the major models and their versions (Sep 2026)



One panel per company; inside each, that company’s current versions. Data: notes/research/r27 — refresh quarterly.

THE MODEL SPECTRUM — WHO SELLS WHICH SPEED (AS OF SEPTEMBER 2026)

Assistant	Fast ↔ thinking, in its own names	Worth knowing
ChatGPT OpenAI	GPT-5.6 family (tiers Sol · Terra · Luna) — one family, speed by setting; the “Think” button (free) or the thinking slider (paid) moves it between speeds; deep research on top.	Largest user base; fast model churn and renaming — tier names rotate.
Claude Anthropic	A ladder of models: Haiku = the fast tier · Sonnet 5 = the everyday workhorse (1M-token context) · Opus 5 / Fable 5 = the frontier deep end — with “extended thinking” as the toggle on top.	Tops coding and real-work evals (SWE-bench, GDPval); 2026 writing evals rank its prose #1.
Gemini Google	Gemini 3.6 Flash = the fast line · Gemini 3 / 3.1 Pro = the reasoning line · Deep Think = the separate hard-reasoning mode (Ultra tier).	1B+ monthly users; the default AI across Workspace (Gmail, Docs, Meet).
Copilot Microsoft	No frontier models of its own — GPT-5.6 plus selectable Claude, inside your tenant. Quick response ↔ Think Deeper; Researcher is its deep-research agent.	The governed option — lives inside the M365 compliance boundary IT already audits.
Perplexity	Routes by question: its own search-tuned stack for fast, cited answers · a “Model Council” of frontier models (GPT + Claude + Gemini) for depth.	Research with receipts — inline citations by default; click through before quoting.
Grok xAI	Grok 4.6, one family — the modes are the spectrum: Auto · Fast · Expert (reasoning) · Heavy (multi-agent, top tier).	Fewer guardrails by design — mind brand-sensitive work.
DeepSeek	The open-weights house, heir of the famous R1: V4 Pro = the reasoning line · V4.1 Flash = the fast line. MIT-licensed — IT can self-host it.	The value leader, ~8 months behind the frontier (NIST CAISI, May 2026); the hosted app stores data in the PRC.

The names above are the September 2026 snapshot and will churn — refresh quarterly. The pattern won't: every vendor sells the same two speeds under different labels. Learn to recognize the tier, not memorize the name.

PART 11 — TEN THINGS WORTH REMEMBERING

The whole course in ten lines

- 1 The prompt is the whole steering wheel — everything the model knows about your task must be in it (or in files it can read).
- 2 Anatomy beats inspiration: Role · Task · Context · Format · Examples — plus the Out and the Stop. A prompt is a requirement, new audience.
- 3 The context window is a desk, not a filing cabinet: documents at the top, question at the end, fresh chat per topic.
- 4 Numbers come from code execution, quotes come from documents, current facts come from search — never from free recall.
- 5 Fluency is not evidence. Uncited claims are drafts. Give the model an out and demand citations.
- 6 Never reveal your preferred answer when asking for judgment — AI affirms you ~49% more than a human would.
- 7 On thinking models: drop the step-by-step scripts, keep the clarity. Contradictions now cost real money.
- 8 A generative prompt says what to write; an agentic prompt is a work order: mission, environment, checks, gates, autonomy.
- 9 Gates catch errors cheapest: definitions before data, reconciliation before analysis, headlines before rendering.
- 10 Prompts that work are assets: name them, version them, store them where the team (and the agent) can find them.

And if you keep only one sentence: a prompt deletes wrong guesses, binds the job, and decides what sits on the desk.

APPENDIX — GLOSSARY: TWENTY TERMS THAT MATTER

Escalation ladder — prompt first, retrieve second, fine-tune last — capability triage

Promotion ladder — one-off → personal → team → packaged → as-code — where a prompt lives

Token — the text chunk a model actually reads (~¾ of a word); pricing and limits count these

Context window — working memory per conversation — everything must fit; cleared when the chat ends

Knowledge cutoff — where training data stops; anything after needs search or your documents

RAG — retrieval-augmented generation — fetch relevant passages, answer from them, with citations

Embedding — a text's coordinate in meaning-space; powers semantic search and RAG retrieval

Hallucination — fluent, confident, wrong — plausibility optimized instead of truth

Few-shot — teaching by 3-5 examples in the prompt

Reasoning model — drafts and checks internally before answering; slower, costlier, better at hard problems

Chain-of-thought — prompting visible step-by-step reasoning — now built into thinking models

Metaprompting — asking the model to critique or rewrite your prompt

Sycophancy — the trained tendency to agree with you; blind the review to defuse it

Agent — model + tools + instructions in a loop, acting until done or stopped

MCP — Model Context Protocol — the open standard for plugging tools into agents

Mission brief — an agentic prompt: goal, environment, inputs, process, checks, gates, reporting

Gate — a human checkpoint inside an agent's process — approve before it proceeds

Open weights — a downloadable model you can self-host — the data-privacy end of the spectrum

Taxonomy — the element → attribute → option catalog behind the Template Creator — your parts list

Context engineering — choosing everything on the desk — prompt, files, history, tool results; parent of agentic prompting